



Week 9 Lecture 5

▼ Class	BSCCS2003
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🔗 Materials	
# Module #	52
▼ Type	Lecture
# Week #	9

Logging

What is logging?

- Record all accesses to app
- Why?
 - Record bugs
 - Number of visits, usage patterns
 - Most popular links
 - Site optimization
 - Security checks
- How?
 - Built into the app → output to a log file
 - Direct output to analysis pipeline

Server logging

- Built in to Apache, Nginx, ...
- Just accesses and URL accessed
- Can indicate possible security attacks →
 - Large number of requests in short duration
 - Requests with "malformed" URLs

- Repeated requests to unusual endpoints

Application level logging

- Python logging framework
 - Output to a file, other "stream" handlers
- Details of application access
 - Which controllers
 - What data models
 - Possible security issues
- All server errors

```

base ~/g/m/gradebook flask run
* Serving Flask app 'application:app' (lazy loading)
* Environment: development
* Debug mode: on
* Running on http://127.0.0.1:5000/ (Press CTRL+C to quit)
* Restarting with stat
* Debugger is active!
* Debugger PIN: 674-210-362
127.0.0.1 - - [06/Sep/2021 21:04:21] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [06/Sep/2021 21:04:21] "GET /static/css/style.css HTTP/1.1" 304 -
127.0.0.1 - - [06/Sep/2021 21:04:21] "GET /favicon.ico HTTP/1.1" 404 -
127.0.0.1 - - [06/Sep/2021 21:04:27] "GET /user/ HTTP/1.1" 200 -
127.0.0.1 - - [06/Sep/2021 21:04:27] "GET /static/css/style.css HTTP/1.1" 304 -
127.0.0.1 - - [06/Sep/2021 21:04:34] "GET /user/1 HTTP/1.1" 200 -
127.0.0.1 - - [06/Sep/2021 21:04:34] "GET /static/css/style.css HTTP/1.1" 304 -

```

Log rotation

- High volume logs → mostly written, less analysis
- Cannot store indefinitely
 - Delete old entries
- Rotation →
 - Keep last N files
 - Delete oldest file
 - Rename log.i to log.i+1
 - Fixed space used on server

Logs on custom app engines

- Google App Engine
 - custom logs
 - custom reports
- Automatic security analysis

Time series analysis

- Logs are usually associated with timestamps
- Time series analysis →
 - How many events per unit time
 - Time of specific incident(s)
 - Detect patterns (periodic spikes, sudden increase in load)
- Time-series databases
 - RRDTOol, InfluxDB, Prometheus, ...

- Analysis and visualization engines

Summary

Security is the key to successful applications

- Requires good understanding of principles
 - Crypto
 - SQL, OS Vulnerabilities, ...
- Good frameworks to be preferred
- Analyze, Identify, Fix